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SUPERSEDING
MIL-B-62152(AT)
2 October 1972

MILITARY SPECIFICATION

BUS, MOTOR: AMBULANCE, INTERCITY, 40 PASSENGER, 6x2 or 6x4

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a six-wheel, four- or two-rear wheel drive, diesel engine driven, intercity type motor bus ambulance, having a seating capacity for 40 adult passengers for long distance transportation. The bus has features to allow ready conversion to accommodate 18 litter patients, the driver, and four seated attendants. Except for customized ambulance conversion features, vehicles procured under this specification are commercial items. Frame and all chassis components, heating and air conditioning equipment shall be commercially available items. The vehicle shall be warranted as specified in the procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2310

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SPECIFICATIONS

FEDERAL

L-S-300	- Sheet and Tape, Reflective: Nonexposed Lens.
F-F-351	- Filters and Filter Elements, Fluid, Pressure, Lubricating Oil, Bypass and Full Flow.
ZZ-T-381	- Tires, Pneumatic, Vehicular (Highway).
L-P-383	- Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
TT-C-520	- Coating Compound, Bituminous, Solvent Type, Underbody, (for Motor Vehicles).
O-E-915	- Extinguishers, Fire, Dry Chemical, Portable.
W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

MIL-P-8013	- Plastic Materials, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
MIL-J-20349	- Jugs, Vacuum or Insulated, Thermal Efficient.

STANDARDS

FEDERAL

FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification, Marking and Data Plate Standards.
MS51317	- Light, Warning, Vehicular, Rotating, D.C.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

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ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
Standards (ANSI, AS and USAS)

ANSI SI.4.71	- American National Standard Specification for Sound Level Meters.
SI.11-66	- American Standard Specification for Octave, Half Octave, and Third Octave Band Filter Sets.
USAS SI.6-67	- USA Standard Preferred Frequencies and Band Numbers for Acoustical Measurements.

(ANSI, AS and USAS Standards are available from the American Standard Institute. Inc., 1430 Broadway, New York, NY 10018.)

SAE, INC.
SAE Standards and Recommended Practices

J366	- Exterior Sound Level for Heavy Trucks and Buses.
J381	- Windshield Defrosting Systems Test Procedure - Trucks, Buses, and Multi-purpose Vehicles.
J382	- Windshield Defrosting Systems Performance Requirements - Trucks, Buses, and Multipurpose Vehicles.

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J537	- Storage Batteries (DoD adopted).
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J688	- Truck Ability Prediction Procedure (DoD adopted).
J816	- Engine Test Code - Spark Ignition and Diesel.
J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA
Vehicle Code of California.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

THE TIRE AND RIM ASSOCIATION, INC .
Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.9.1, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special

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bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, and data plates shall be in accordance with MIL-STD-1223, except the vehicle exterior color shall be gloss white matching color chip No. 17875 of FED. STD. No. 595 and additional markings identifying the vehicle as an ambulance shall be furnished as follows:

(1) The word "AMBULANCE" in reflective color black shall be applied as follows using marking material specified in MIL-STD-1223:
   (a) Front: 1-1/2 inch high letters, centered below windshield.
   (b) Sides: 3-inch high letters, centered under cross and below red stripe.
   (c) Rear: 3-inch high letters, centered across the rear on the lift platform below the red stripe.
(2) A red belt strip shall be painted completely around the coach body below the main windows, and also around the exterior side of the lift platform in its travel position.
(3) Four red crosses shall be painted: one on the top of the coach at the vehicle mid-section; one on each side of the coach at the mid-section located above the main windows; and one on the upper portion, exterior side of the lift platform in its travel position. Crosses shall be as large as practicable within the limitations of the vehicle configuration.
(4) A 3-inch red cross shall be painted on the vehicle front to the right and left and on the same plane as the word "AMBULANCE."
(5) The belt strip and crosses shall be gloss red matching color chip No. 11136 of FED. STD. No. 595.
(6) The crosses shall be on a square background of reflectorized silver material conforming to type I, Class 3, reflectivity 1 of L-S-300.

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A detailed sketch or drawing depicting vehicle exterior finish and color scheme, vehicle identification markings and ambulance markings shall be submitted to the contracting officer for approval prior to painting.

* 3.1.1.2 Heavy duty cooling system. A heavy duty cooling system shall be furnished that will maintain coolant at a temperature below the boiling point with vehicle loaded to rated gross vehicle weight (GVW) and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 112 degrees Fahrenheit (°F). Thermostatic control of engine coolant temperature shall be provided. Control of coolant temperature shall be accomplished by either thermostatically controlled radiator shutters, a thermostatically controlled variable pitch fan, or by thermostatically controlling the flow of all engine coolant through the radiator.

* 3.1.1.3 Coolant recovery system and level indicator. A radiator coolant recovery system or surge tank of adequate size to prevent loss of coolant under conditions specified in 3.1.1.2 shall be furnished. A translucent reservoir, sight gage or other means for checking the coolant level shall be provided to enable the operator to check the coolant level while standing on the ground. In addition, a low coolant level and a high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

* 3.1.1.4 Drain plugs. Drain plugs installed in the transmission and rear axle(s) shall be of the permanent magnet type.

* 3.1.1.5 Oil filter. A full flow and a bypass type oil filter shall be furnished unless otherwise recommended by the engine manufacturer. The full flow oil filter shall be connected directly to the engine lubrication system and the bypass filter by flexible lines. The inlet and outlet connections of the bypass filter shall be of different sizes to preclude improper assembly. The filtering element of the bypass filter shall be in accordance with F-F-351.

* 3.1.1.6 Wood treatment. Wood flooring shall be treated in accordance with MIL-STD-1223 or with the manufacturer's standard commercial practice providing equivalent protection.

* 3.1.1.7 Towing device(s). Towing device(s) for towing the bus shall be provided on the front of the bus. The towing device(s) and the front of the bus shall withstand, without any sign of failure, turning the bus within its minimum turning radius and backing the bus on a level roadway.

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* 3.1.1.8 Warning lights. One warning light conforming to MS51317 shall be mounted in the front center of the bus on the roof section immediately above the driver's compartment, and one on the rear of the bus, center or right of center. Neither light shall extend above the roof line of the coach main body, and each shall be properly reinforced at points of attachment. The color of warning lights shall be red. Both lights shall be operable by means of a separate switch with an indicating light installed on the driver's instrument panel. The switch and indicating light shall be properly identified.

3.1.1.9 Ambulance conversion features. The vehicle shall be provided with various ambulance conversion features specified herein.

3.1.1.9.1 Drawings. Overall outline sketch(es) or drawing(s) shall be furnished with the supplier's proposal or quotation showing location of doors, seating, equipment, litter installation and lift platform mechanism. Prior to start of production the contractor shall furnish two sets of complete layout and general assembly drawings of the proposed vehicle to the engineering office of the procuring activity for customer approval. After approval, any modifications shall be likewise submitted for customer approval.

3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

* 3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Interstate Motor Carrier Noise Emission Standards when tested in accordance with the Regulations of the Department of Transportation, Part 325.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis, body with all attachments, accessories, litter kits, and equipment furnished; full complement of fuel, lubricants and coolant; and operator (weight computed at 175 pounds).

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3.2.5 Gross vehicle weight. The gross vehicle weight (GVW) shall be curb the combined weight and payload weight. The payload shall be the weight of 40 passengers computed at 175 pounds per passenger plus 25 pounds baggage allowance per passenger.

* 3.2.6 Dimensions. Overall bus dimensions shall be as follows:
(a) Overall width shall not exceed 96 inches.
(b) Overall height shall be not greater than 138 inches.
(c) Overall length shall be not greater than 40 feet.
(d) Ground clearance shall be not less than 7-3/4 inches under axles and not less than 10 inches under all other parts.

3.2.7 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit ready accessibility for routine servicing and shall permit accessibility for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

3.3 Performance.

* 3.3.1 Speeds. High and low speed requirements shall be met with the bus loaded to specified GVW.

* 3.3.1.1 High speed gradeability. The bus shall ascend a 0.5 percent grade at 60 mph. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3). When air conditioning is specified and powered by the main engine, the full load of the air conditioner shall be deducted in gradeability calculations.

* 3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed not greater than 6.5 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 68 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3)}}$$

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* 3.3.2 Braking. The service brakes shall control and hold the vehicle when loaded to its specified GVW, on a 30 percent grade. Under all conditions of loading, the service brake shall stop the bus, from a speed of 20 mph, within a distance of 30 feet from the point at which movement of the brake pedal begins. Deceleration and stopping distance shall be demonstrated on dry, hard, approximately level road surface that is free from loose material.

* 3.3.2.1 Downhill braking. The vehicle braking system shall include at least one of the following features:

(a) Brakes with a total of not less than 1,930 square inches of effective brake area
(b) Automatic transmission with built-in hydraulic retarder
(c) An engine type brake that brakes by engine compression through opening exhaust valves near piston top center and venting to the atmosphere. The engine brake shall include modulated control. The engine brake shall permit the bus loaded to GVW on a dry, hard road surface to negotiate a six percent downgrade without exceeding 27 mph, within the operating restriction of the engine and transmission and without the use of service brakes.

3.4 Chassis.

* 3.4.1 Engine. The bus shall be furnished with a liquid cooled compression ignition, two-stroke or four-stroke cycle diesel engine. The engine shall have not less than six cylinders. The engine shall be the manufacturer's standard or optional engine for the commercial model vehicle which meets or exceeds the requirements of this specification. The engine shall be mounted at the rear or mid-mounted under the floor between the front and rear axles. An engine governor set and sealed to limit the engine speed to the maximum recommended operating speed shall be provided. An oil cooler shall be furnished. Engine net horsepower used in performance prediction calculations shall be determined in accordance with SAE J816 or SAE J1349.

* 3.4.1.1 Engine compartment. The engine compartment ventilation and design shall allow operation without damage to engine, transmission and all components when the vehicle is operated in ambient temperatures of up to 112°F. Ready access to engine compartment shall be provided for servicing and routine maintenance of engine and engine components. Hinged engine compartment covers or doors shall be provided with a device to hold them open for servicing and maintenance.

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* 3.4.2 Starting system. The bus shall be furnished with a 12- or 24-volt starting system. Engine starting equipment shall include a glow plug or a pressurized ether system. The ether system shall include all equipment, including container mounting bracket, and any necessary piping. The ether system shall be of the automatic injection, temperature-limited type, controlled from the driver's area, inoperative with the engine warm; or shall be of the measured shot type, manually or key operated from the driver's compartment, with a reservoir, or provisions for a reservoir, of not less than 12 fluid ounces, inoperative with either the engine warm or the engine running.

* 3.4.2.1 Starter system controls. The starting motor shall be switch-operated by current fed in parallel from a toggle switch in the engine compartment to both a “start” switch on the instrument panel and a “start” switch in the engine compartment and then to the operating coil of the relay. A safety switch shall be provided in the engine compartment so that the engine cannot be cranked from the driver's position when this switch is in the “OFF” position. Means shall be provided to prohibit cranking of the engine except when the transmission is in the neutral position.

* 3.4.3 Alternator. When air conditioning is furnished and powered by the main engine, the alternator shall provide not less than 120 amperes at engine idle speed, and not less than 200 amperes at not more than 60 percent of the engine governed speed for a 12-volt system; or not less than 60 amperes at engine idle speed, and not less than 100 amperes at not more than 60 percent of the engine governed speed for a 24-volt system.

* 3.4.3.1 Independent 120-volt alternator. When an independent air conditioning system is provided with a separate 120-volt alternating current (ac) alternator driven by the auxiliary engine, the main alternator shall have a direct current (dc) output of not less than 90 amperes at engine idle, 125 amperes at 60 percent of engine governed speed, and 140 amperes rated capacity. When a 120-volt alternator is furnished, it shall have adequate power for operation of all air conditioning fans and blowers and shall have a rated output of not less than 1,200 watts at 120 volts ac, single phase, 60 Hertz.

* 3.4.4 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67 and shall provide all necessary piping, fittings, and accessories.

* 3.4.4.1 Air cleaner. An air cleaner(s) shall be furnished. The air cleaner(s) shall be of the dry type, with air restriction indicator to show when the filter(s) should be replaced.

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* 3.4.4.2 Fuel tanks. A fuel capacity of not less than 125 gallons shall be provided. When more than one tank is furnished, a balance line shall be provided. The filler shall be identified and arranged for safe and convenient filling from the exterior of the body. The access for the filler neck shall be located to prevent any accumulation of spilled fuel between the body panels or shall be sealed to prevent entry of spilled fuel to any point inside the exterior body panels.

* 3.4.5 Fuel tank protection plate. If the fuel tank extends below the chassis frame, or body skirt, the tank shall be protected on the underside and at both ends by a steel plate of not less than 12 gage (0.1046 inch) thickness. The protection plate shall provide for drainage of any accumulated liquids and shall permit access to the fuel tank drain plug. A fuel tank protection plate is not required if a Federal Motor Carrier Safety Tank is provided.

* 3.4.6 Transmission. Transmission input torque capacity shall be at least equal to the maximum torque delivered by the engine.

* 3.4.6.1 Automatic transmission. Unless otherwise specified, the manufacturer's standard automatic transmission recommended for the make and model coach shall be provided. The transmission shall provide required performance under all normal operating conditions and under all conditions specified herein. The transmission shall include a torque converter.

* 3.4.6.2 Manual transmission. When specified (see 6.2), the manufacturer's standard constant mesh manually shifted transmission shall be furnished. The transmission shall provide at least four forward and one reverse speeds.

* 3.4.6.2.1 Clutch. The clutch shall have a torque capacity exceeding the maximum delivered engine torque and shall be the largest capacity clutch offered for the engine and the vehicle model furnished. Pedal effort required to disengage the clutch shall not exceed 60 pounds. Spring or air assists, when employed, may be attached or incorporated within the clutch assembly.

* 3.4.7 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction. Propeller shaft guards shall be provided in accordance with Federal Motor Carrier Safety Regulation 393.89.

* 3.4.8 Chassis components. Chassis components shall be designed for bus construction. The body and the chassis shall be of the integral type. Body-on-chassis type construction, bolted or welded, is not acceptable.

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* 3.4.9 Suspension. The vehicle shall be equipped with a full air suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to GVW as specified in 3.2.5. The vehicle shall be provided with hydraulic double-acting shock absorbers: 2 on the front and 4 on the rear drive axle(s). Shock absorbers shall also be provided at the rear non-drive axle, if manufacturer's recommended practice. The suspension design shall be such that the natural frequency of vertical vibration about any transverse axis shall be less than 1.5 cycles per second at rated capacity for specified GVW. Outside rear wheels shall be designed and located so that there will be no interference with the use of tire chains. The air system shall incorporate leveling valves with a time delay arranged to avoid constant air consumption due to road shocks. The suspension shall be provided with a mechanism at each wheel to assure lifting of the wheel and axle when jacking the vehicle from the applicable jacking location (see 3.5.2).

* 3.4.9.1 Air springs. The front axle shall be equipped with four air springs of 8 inches minimum diameter, or two air springs of 12-1/2 inches minimum diameter. Each rear drive axle shall be equipped with four air springs of 10 inches minimum diameter, and non-drive rear axle shall be provided with the manufacturer's standard air springs. All axles shall have none of the sprung weight carried on anything other than air springs. No other spring medium such as leaf, coil or rubber springs shall be used in conjunction with air springs.

3.4.9.2 Radius rods and sway bars.

* 3.4.9.2.1 Radius rods. The front axle shall be provided with not less than four radius rods to control lateral and longitudinal movement. Radius rods shall transmit driving and braking forces to vehicle chassis and body. The rear axle tandem assembly shall be provided with not less than eight radius rods: four to control longitudinal movement and four to control lateral movement; or the drive axle shall have four radius rods to control lateral and longitudinal movement. The non-drive axle shall be of an independent or swing arm suspension design incorporating adequate means to control lateral and longitudinal loads such as upper and lower “A” frames and an adjustable toe-in link. Each end of the radius rods shall be equipped with rubber bushings that do not require periodic lubrication.

* 3.4.9.2.2 Sway bars. Sway bars, of adequate size and strength, shall be incorporated in both front and rear suspension systems. Transverse front and wish-bone type radius rods at the rear may be provided for lateral stability.

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* 3.4.10 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to GVW as specified in 3.2.5. The vehicle shall be provided with two rear axles, one or both being drive axles. Single reduction drive axle(s) shall be furnished. When specified (see 6.2), oil seals shall be provided, on all axles.

* 3.4.10.1 Traction control. When specified (see 6.2), 6x2 vehicles shall be furnished with a traction control on the drive axle. The traction control shall actuate automatically to insure that power is transmitted to the wheels having traction when the opposite wheel loses traction. When a 6x4 vehicle is furnished, the dual drive axles shall incorporate a means of locking out all differential action, providing full traction on all rear wheels. The locking and unlocking device shall be operated by a switch within easy reach of the driver in the normal seated position. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.11 Wheels, rims and tires. The vehicle shall be equipped with single wheels on the front axle and with dual wheels on the rear drive axle(s). Tire size, tread design and load range (ply rating) shall be the same for all tires on the front axle and rear drive axle(s). When a non-drive rear axle is furnished, dual or single wheels may be furnished. Tire and rim size may be smaller than those furnished for the drive axle, but shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the vehicle loaded to the GVW specified in 3.2.5. Rims and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.11.1 Tires. Tires shall be tubeless steel belted radial type with highway tread. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to the GVW specified in 3.2.5. When a non-drive axle is furnished and the tires supplied on the non-drive axle are smaller than those on the drive axle, the tires supplied on the non-drive axle are not required to be steel belted radial design. Tires shall conform to Tire and Rim Association recommendations or to ZZ-T-381 with a size designation system the same as the Tire and Rim Association.

* 3.4.11.2 Balancing. Each tire shall be balanced within practicable limits. Wheels, hubs, and brake drivers shall be effectively in balance. Balancing shall be adequate to preclude wheel shimmy at all vehicle speeds.

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* 3.4.11.3 Spare wheel assembly. Unless otherwise specified (see 6.2), a spare wheel or rim, for the front and for the rear drive axle(s), shall be mounted on a carrier and installed in a readily accessible location. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.18. The carrier and spare shall not be located in the baggage compartments. The spare wheel may be stowed in a suitable separate compartment.

* 3.4.11.4 Spare tire. Unless otherwise specified (see 6.2), an inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread design, and load range as the tires in use on the front axle and the rear drive axle(s).

* 3.4.12 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42(b) and 393.44 through 393.52.

* 3.4.12.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

(a) Air compressor, unloader-head-type, engine-driven and engine-lubricated, air or water-cooled, and having a capacity of not less than 13 cubic feet per minute (cfm)
(b) Air storage reservoir(s) with a total capacity of not less than 4,200 cubic inches. Each tank shall be equipped with drain, and with safety and check valves between compressor and last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Alcohol aspirator, with unbreakable transparent container
(i) Automatic moisture ejector
(j) Ping tank or discharge muffler
(k) Automatic slack adjusters.

NOTE: When an air dryer is furnished between the air compressor and the tank, (h) the alcohol aspirator and (i) the automatic moisture ejector shall be omitted.

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* 3.4.13 Electrical system. A 12-volt, 24-volt, or combination 12-volt 24-volt electrical system shall be furnished. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.13.1 Lighting. All vehicle lights, reflectors and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.23 through 393.26. Lights and reflectors shall not be mounted on rub rails (unless recessed and fully protected) or the vehicle bumpers. An illuminated rear license plate holder shall be provided. Engine compartment illumination consisting of four light fixtures connected to one switch shall be provided. Fixtures shall be placed so as to provide uniform compartment illumination.

* 3.4.13.2 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall be provided with single-faced front signal units and single-faced rear signal units installed in accordance with SAE J588. Turn signal operating units shall be self-cancelling type with visible and audible flash indicator. An additional set of armored type amber turn signals, mounted on each side of the bus above the front wheel wells, shall be furnished.

3.4.13.3 Interior lighting. Interior lighting shall include overhead lights for general illumination, reading lights and litter lights. Fluorescent light fixtures providing equivalent illumination may be used in lieu of bulb-type fixtures. Fluorescent fixtures, when used, shall be equipped with radio interference suppression devices. Master control on-off switches to simultaneously control the interior lighting shall be provided at both the front and the rear of the vehicle for use by the driver and the attendants, respectively. Separate master switches shall provide on-off control of the entire reading and overhead lighting circuits and entire between-tier lighting circuit.

3.4.13.3.1 General illumination. General illumination shall be provided by 12 light fixtures mounted in the ceiling or overhead duct. Six of the fixtures shall operate from both 12-volt and 110-volt supply and six shall operate from 110-volt supply only. Three-way on-off switches shall be provided at the front and rear of vehicle for the 12 fixtures.

3.4.13.3.2 Reading lights. Individually controlled reading lights shall operate from 12-volt dc and shall be installed in the ceiling over each seat, with all seats in place for the 40 passenger configuration. With all reading lights on, the illumination on an eleven by sixteen

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inch plane at 40 degrees to the floor, with the lower edge at 33 inches from the floor at the front edge of the seat, shall average not less than 20 foot candles. Reading lights shall be installed in such a manner as to provide specified lighting when litter stanchions are in the stowed position. Reading lights shall be remotely controlled by switches located in the sidewall between each row of seats. Switches shall be readily accessible to each seated passenger.

3.4.13.3.3 Litter lights. Twelve portable litter lights shall be furnished with each vehicle. The litter lights shall be individually controlled, shall operate on 12 volts dc, and shall incorporate on-off switch in the light body or the light cord. The litter lights shall provide suitable illumination between tiers of litters and above the upper tiers of litters for use by attendants in the care of patients. (The reading lights will also illuminate the upper tiers of litters.) The litter lights shall be designed to shield the attendant and patient from direct glare, and shall have provisions for attachment to the litter support posts. Each light shall be swivel mounted for ease of adjustment and shall have a standard two-prong electrical plug for connection to the 12-volt dc power supply.

3.4.13.3.4 Blue bulbs. Blue bulbs shall be furnished at not less than two interior locations and shall be connected to and operated by the marker light circuit.

3.4.13.3.5 Driver and door lights, 12-volt. A domelight, actuated both by opening of the service doors or by the driver, shall be provided for illumination of the driver area. Door-operated step lamps shall be installed to provide illumination of the stepwell and the ground adjacent to step.

3.4.13.4 Floodlight. A 110-volt shockproof floodlight shall be mounted at the rear of the coach above the center of the rear loading door. The floodlight shall be securely mounted, shall be adjustable and shall include a locking device to secure it in any preselected position. The floodlight shall be capable of illuminating at least the area required for operation and occupation of the rear lift platform. The floodlight shall not extend above the roof line of the main coach body, nor beyond the lift platform when folded against the rear of the coach.

* 3.4.13.5 Electrical power. Vehicle on-board electrical power shall be adequate for all vehicle equipment. In addition to the 12-volt dc alternator driven by the main engine and the batteries (see 3.4.3 and 3.4.14), the vehicle on-board electrical power shall include:

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(a) A permanent magnet (PM) type engine driven alternator rated at not less than 1200 watts at 110 volts single phase, 60 Hertz.
(b) A power pack device or devices, Lorain Model WBA-601M, or equivalent, capable of converting the 12-volt dc vehicle battery power to provide not less than 300 watts at 110 volts, single phase 60 Hertz to wire molds, located on each side of the bus. The power pack shall be capable of handling both inductive (induction motor) and resistive loads without damage to loads or the power pack. A warning light shall be furnished adjacent to the disconnect switch to indicate when the switch is in the "on" position to prevent excessive battery drain.

The vehicle electrical system shall also include a 110-volt male outlet mounted on the exterior curbside at the rear, for connecting an exterior power source to the vehicle. A transformer and rectifier shall be provided to reduce a portion of the 110-volt, 60 Hertz, single phase shore power to 12 volts dc for use on appropriate 12-volt circuits. Protective devices and line disconnects/ switches shall be provided to select adequate vehicle power or shore power for the 12-volt dc and 110-volt ac components. Component voltages and source of power for the components for both parked-standby and over-the-road operations shall be as indicated in table I.

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TABLE I. Electrical components and power sources.

Components			Power sources*							
			Parked standby operation (Use both engines when needed)				Over-the-road operation			
			Without shore line		With 110-volt shore line		Main engine only		Both engines	
(Voltages)	12	110	12	110	12	110	12	110	12	110
Lift gate	X	-	1	-	1	-	1	-	-	-
Gen. illum.	X	X	-	3	-	5	1	-	1	3
Reading lights	X	-	-	-	-	-	1	-	1	-
Litter lights	X	-	1	-	4	-	1	-	1	-
Blowers (HV A/C)	X	-	-	-	4	-	1	-	1	-
Aux. heater	X	-	-	-	4	-	1	-	1	-
Pumps (water)	X	-	1	-	4	-	1	-	1	-
Condenser (A/C)	X	-	-	-	4	-	-	-	3	-
Lav power vent	X	-	-	-	4	-	1	-	1	-
Wire mold	X	X	1	2	4	5	1	2	1	2
Vac pump	-	X	-	3	-	5	-	-	-	3
Floodlight	-	X	-	3	-	5	-	-	-	-
Bus engine/chassis	X	-	1	-	1	-	1	-	1	-
Bus exterior lights	X	-	1	-	1	-	1	-	1	-

*NOTE: Power sources are:

#1 Main engine batteries and alternator (12-vdc).
#2 Inverter connected to main engine batteries (110-vac).
#3 PM alternator driven by engine (110-vac).
#4 Transformer/rectifier connected to 110 shore line (12-vdc).
#5 Shore line (110-vac).

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3.4.13.6 Electrical outlets. A heavy duty wire mold shall be installed on each side of the coach extending between the rearmost and foremost litter support poles. Each wire mold to provide both 110-volt and 12-volt outlets as follows: two duplex 110-volt outlets and one duplex 12-volt outlet at each end, and three duplex 110-volt outlets plus one duplex 12-volt outlet between each bank of litters for a total of ten duplex 110-volt outlets and four duplex 12-volt outlets on each side of the coach. The 110-volt outlets shall be one-prong 2 blade #115252 grounded Hubbel duplex or equal. The 12-volt outlets shall be twistlok.

3.4.13.7 Wiring. All wiring shall provide adequate electrical insulation, shall be waterproof, and where required, shall provide adequate flexibility and resistance against solvents or abrasion. All wiring in the engine compartment shall be covered with heat and oil resistant type covering. Wiring shall be color coded or number coded so that the circuit of each wire may be readily determined by observation of any 3-inch segment along the wire. All main wiring harness shall be concealed within body and secured in position by rubber covered clamps with rolled edges.

3.4.13.8 Wiring installation. All wiring shall be properly grouped and shall be so installed as to permit ease of replacement. It shall not be run through metal or other parts of the structure, except where unavoidable and at such points, suitable rubber bushings shall be provided. Every precaution shall be taken to avoid damage from heat, water, solvents, or chafing. Loom may be used provided it meets the above conditions except in the engine compartment.

3.4.13.9 Size of conductors. All conductors and terminals shall be of proper size to carry the maximum current load in the circuit with a maximum drop of 1/2-volt, except in the starter circuit.

3.4.13.10 Electric circuit protection. Fuse panels, where provided, shall be of rugged mechanical construction. Fuse clips shall be rubber covered where provided, and have the acceptable margin of electrical and mechanical safety to prevent heating of the clips in normal service, distortion during installation and removal of fuses. Where practical, automatic circuit breakers of ample capacity shall be used. Where wiring passes through plates or other structural members, grommets of equally adequate protection for the insulation shall be provided.

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3.4.13.11 Wiring connections. No splicing of wires shall be permitted. Wherever connections are required they shall be made by means of soldered or special non-soldered terminals of carrying capacity equal to or greater than the wire itself and equipped with substantial and non-corroding means for attachment.

* 3.4.14 Batteries. Batteries shall be 12- or 24-volt potential. Total capacity of all batteries shall provide not less than 400 ampere-hours for 12 volts and 200 ampere-hours for 24 volts at a 20-hour rate. Batteries shall be accessible for inspection without detaching the batteries from the vehicle. The batteries may be mounted in the engine compartment or on trays or the battery compartment floor. The battery tray or battery compartment floor shall be acid resistant stainless steel or be coated with acid resistant paint. When batteries are mounted in a special battery compartment, the compartment shall be large enough to inspect and service the batteries without requiring their removal from the coach. A battery access door shall be provided on the side of the body. Access to the batteries through the baggage compartment door is acceptable. If battery trays are used for inspecting the batteries, they shall swing out or slide out on rollers, if necessary, to support the batteries. The batteries shall be located away from or protected from engine/transmission heat that will have an adverse effect on battery life or operation.

* 3.4.14.1 Battery cables. Battery cables shall be routed and secured in such a manner as to preclude chafing the cables at any point. Insulated studs shall be used to pass through the frame and the bulkheads. Cables shall be so mounted, fastened, and protected that they will not become loose during service. The edges of all holes in metal members through which the cables pass shall be bushed with rubber grommets.

* 3.4.15 Steering. Power steering shall be furnished. Power steering tubing, fittings, and mechanical connections shall be located to provide easy access for inspection and servicing.

* 3.4.16 Windshield wipers and washers. The vehicle shall be equipped with dual, air operated windshield wipers of the parallelogram type. The wiper system furnished shall deliver 33 pound feet of torque and sweep the maximum practicable area of the windshield surface. Windshield wiper arm tip force shall be not less than 36 ounces. Air operated windshield wipers motors shall be "Sprague Super Challenger" or equivalent with 5/8-inch diameter drive. Wipers shall conform to Federal Motor Carrier Safety Regulation 393.78. Windshield washers shall be provided. Washers shall be electric or air operated with a control on the instrument panel.

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* 3.4.17 Bumpers. Front and rear bumpers shall be provided. The bumpers shall be of anodized aluminum section, stainless steel, or chrome-plated steel. The rear bumper shall conform to Federal Motor Carrier Safety Regulation 393.86 and shall be as specified in 3.5.11.6.

* 3.4.18 Tools. Unless otherwise specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle, and wheelnut wrench. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly. When jack height exceeds the height under the jacking location with the tire flat, run-up blocks shall be furnished. Storage provisions for the tool equipment and tire chains shall be furnished. The stowage provisions shall be accessibly located and weatherproof.

* 3.4.19 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's seat. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be provided with indirect illumination. The instrument panel shall be designed to preclude windshield reflection of instruments.

* 3.4.20 Instrumentation. The vehicle shall be complete with at least all the controls, indicators, and gages furnished as standard equipment by the manufacturer. All gages shall be true and accurate. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Voltmeter, indicating extent of battery charge or discharge
(c) Fuel gage
(d) Speedometer
(e) Recording odometer or hubodometer
(f) Oil pressure gage with visible and audible warning for low oil pressure
(g) Engine coolant temperature gage with visible and audible warning for overheated engine
(h) Indicator light to show headlight high-beam operating
(i) Indicator light to show operation of stoplights
(j) Turn signal pilot indicator
(k) Transmission high temperature gage, when an automatic transmission with a retarder is furnished

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(l) Low transmission oil pressure warning light and buzzer when a manual transmission with a pressurized oil system is furnished.
(m) Tachometer

3.4.20.1 Horns. An electric and an air-operated horn shall be provided.

3.4.21 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.4.22 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust tailpipe shall extend to the rear bumper and shall be so located as to minimize the possibility of exhaust fumes entering the body. The exhaust system beyond the engine compartment shall be insulated from the fuel tank and tank connections by securely attached metal shields at any point where it is 12 inches or less from the tank, tank connections, or any fuel line connections.

3.5 Body. The bus shall provide seating for 40 adult passengers. The bus shall be customized to provide ambulance features specified herein for ready conversion to accommodate 18 litter patients plus four seated attendants. The body shall have full-skirted sides, shall present a generally symmetrical appearance, shall exclude engine fumes and exhaust gases from the bus interior, and shall provide safe and comfortable transportation for passengers and driver.

3.5.1 Inside body dimensions. At seat level, body inside width shall be the maximum reasonably attainable within the maximum permissible overall vehicle width of 96 inches. Between the first and last vertical roof bows through center of aisle, inside height shall be not less than 79 inches. To allow movement of litter bearers, the aisle width, with the litters installed, shall be not less than 38-1/2 inches.

* 3.5.2 Construction. The body and the chassis shall be of integral-type construction. The body and the roof framing shall be constructed of metal sections with insulated aluminum skin. All parts, where applicable, shall be fastened together in a manner which will preclude loosening of bolts, screws, rivets and cracking of welded joints when the vehicle is operated under intended service condition. Jacking locations shall be provided at each wheel. Plates shall be of sufficient capacity to withstand the jacking force required to lift the portion of the loaded vehicle affected without damage to any vehicle component. Self-tapping screws shall not be used where

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subjected to stress. Wood shall not be used for structural framing. Outside body panels and other sheet metal shall be of aluminum alloy or stainless steel of not less than the gages, as applicable, regularly furnished by the manufacturer in a bus of the model offered. The body, roof, and panel joints shall be watertight and leakproof with leading panels overlapping the edge of trailing panels. The pitch of fasteners (bolts, screws, and rivets) at all major interior and exterior sheet metal panel edges shall not exceed 24 times the thickness of the thickest material used in the joint. The strength of such edge joints shall be equal to not less than 60 percent of the tensile strength of the thinnest member.

3.5.2.1 Steady state noise levels. The vehicle design shall limit noise level to the minimum practicable level. The vehicle shall be tested as specified in section 4, Quality Assurance provisions (see 4.3.4.1). Sound pressure levels shall not exceed the values specified in table II.

TABLE II. Maximum noise levels (Preferred Frequencies USAS S1.6-1967).

Octave band limits (Hz)	Center frequency (Hz)	Noise level (dB)
44 - 87	63	95
87 - 175	125	90
175 - 350	250	85
350 - 700	500	80
700 - 1400	1000	80
1400 - 2800	2000	68
2800 - 5600	4000	65
5600 - 11200	8000	55

In addition, the overall sound level shall not exceed 83 dB(A)

* 3.5.3 Floor. The floor frame and floor plates, with entrance steps and whelhousings, shall be of substantial construction and shall conform to Federal Motor Carrier Safety Regulation 393.84. Wood flooring of not less than 1/2-inch, 5 ply marine grade or Douglas fir improved select sheeting, exterior grade plywood shall be furnished. Wood parts shall be treated as specified in 3.1.1.6. Steel or extruded aluminum channels, ranging in size from not less than 1-1/8 to 2 inches in depth and spaced to provide support for the loads imposed shall be furnished under the floor. Floor shall be of single level rather than step design.

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3.5.3.1 Entrance steps, lift platform or ramp. Access steps shall be provided at the service entrance for walk-on patients. Steps shall be of the standard bus stepwell design. A lift platform shall be furnished at the loading doors specified in 3.5.4.

* 3.5.3.1.1 Stepwell. The stepwell shall be of the enclosed type with two, three or four steps, and constructed of patterned or lined anodized aluminum, stainless steel or fiberglass. The height of the bottom step shall not exceed 16 inches. Riser heights shall provide equal step height within 1/2 inch. Step covering shall be not less than 5/16-inch thick ribbed rubber with white molding, or ribbed stainless steel nosings.

3.5.3.1.2 Lift platform. A hydraulically operated lift platform shall be provided. The platform shall be of adequate size to permit moving a loaded litter into the ambulance through the loading door by a normal litter bearing team without danger of falling from the platform and with one smooth forward motion. The platform shall be capable of being lowered to the ground and raised to ambulance floor level when it is in a horizontal position. It shall also be capable of being raised above horizontal up to 90 degrees and folded flat against the vehicle (for closed position during travel of the ambulance). The platform shall be not less than 118 inches long, with the outboard end ramped for approximately 18 inches at no greater than a 6 degree slope, to provide for ease of loading wheelchairs onto the platform, or for use as a ramp. Removable or folding hand rails shall be provided as needed for safe use of the platform by ambulatory, wheel chair, or litter patients. The platform shall be covered on the walking surface with durable, weather-resistant non-skid material, suitable for use by persons using crutches or canes. The surface shall be level and without ridges or gaps. The platform design shall permit safe locking in all positions where support of individuals is possible and under loads anticipated (for example, litter, patient, and bearers, total 775 pounds). The platform design may include provisions for folding or telescoping the platform for over the road travel. The design of the lift platform and the loading doors shall insure that the platform in the raised horizontal position clears the opened doors in such a manner that it is impossible for a hand or foot to be pinned between them. Except for installation of the hand rails, all lift operations shall be hydraulically controlled. A manual hydraulic pump shall be provided to lower and raise the lift platform in the event of power failure or emergency. The lift controls shall be conveniently located on the interior and exterior at the loading door. Exterior controls shall be weatherproof.

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3.5.3.2 Wheelhousings. Wheelhousings shall include splash shields, and shall provide ample clearance for use of tire chains on the outside duals. Molded rubber or metal splash trim shall be installed around the inside or outside lip of the outside edge of both the front and rear wheelhousings.

3.5.3.3 Floor coverings. Except for the driver area, the complete interior of the bus floor shall be covered with not less than 1/8-inch thick rubber or synthetic rubber padding and washable, flash flame proof carpeting material. Padding and carpeting shall be of top grade, as recommended by the carpet manufacturer for buses, restaurants and heavy traffic areas.

3.5.4 Service and loading doors arrangement. The vehicle shall have a bus-type service door for walk-on patients and a large loading door with lift platform for loading and unloading litter patients, wheelchair patients and ambulatory patients accompanied by attendants. The loading door shall be located at the rear, permitting straight-in movement of litter patients. A bus-type service door shall be provided on curbside towards the front so as not to interfere with the litter racks when the bus is set up for all eighteen litters.

3.5.4.1 Door construction. The service door and the loading door shall have safety glass panel(s), and compression and overlapping seals to exclude dust, water and cold air. The sedan-type service door shall be equipped with a key operated lock. The door operating linkage shall be installed within the body or enclosed to prevent the accumulation of mud, ice, and snow.

3.5.4.2 Service door. The service door shall be of the solid sedan type with manual driver control or shall be of the two-leaf outward opening air operated type. Controls shall be conveniently located for operation by the seated driver. Minimum dimensions of the clear opening, unobstructed by door seals or door opening mechanism, shall be 28 inches by 78 inches for the leaf-type door and shall be not less than 24 inches in width at center, 20 inches at the top, and 74 inches in height for the sedan-type door. The bus shall be equipped with a concealed release to open the service door from the exterior of the bus.

3.5.4.3 Loading door. The minimum clear opening of the loading door shall be 39 inches in width and 74 inches in height. The size and location of the door shall allow continuous straight-in movement of an attendant with a wheelchair and of stretcher bearers. Door latching mechanisms shall be provided inside and outside of the vehicle. Heavy duty retaining devices shall be provided to retain the door in the full open position for loading.

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3.5.4.4 Environmental conditions. The service door and loading door design and installation shall allow operation of doors under all weather conditions including snow and freezing rain, and shall not result in vehicle interior noise levels exceeding those specified in 3.5.2.1 under mobile operating conditions.

* 3.5.5 Sides and roof. The sides and the roof shall have sheet metal outer panels, lined on the inside and integrated with the body structure. The front and rear roof cap sections may be constructed of unstressed molded fiberglass sections in lieu of sheet metal. The vehicle front cowl section and rear end section shall be metal outer panels or fiberglass reinforced thermosetting plastic (see 3.5.5.1.1).

* 3.5.5.1 Exterior. The sides shall be equipped with rub rails. When integral with the side panels, the rub rails shall run at least between the front and rear wheelwells. When rub rails are not integral with the panels, rub rails shall run the full length of each side of the coach. The side panels shall be a combination of fluted stainless steel or fluted anodized aluminum panels and smooth painted aluminum panels. Front and rear panels shall be of the same construction except reinforced plastic may also be used. Contractor shall submit photographs or illustrated brochures depicting type of exterior construction to the requisitioning activity, for approval prior to production.

* 3.5.5.1.1 Fiberglass reinforced plastic front cowl and rear end section. When fiberglass front and rear sections are furnished, the sections shall be not less than 0.210 inch in thickness. When fiberglass front and rear roof cap sections are furnished, the sections shall be not less than 0.120 inch in thickness. The material shall conform to type I of L-P-383 and shall have the minimum mechanical properties for "Mat" fabric listed in table I of MIL-P-8013. A polyester resin and fiberglass depositor may be employed to apply the resin and fiberglass. The plastic sections shall be composed of not less than 40 percent (by weight) glass fibers and additives to the resin mix shall provide for fire-retardant and weathering performance properties. Color additives need not be added to the resin mix.

3.5.5.2 Interior. The body interior shall be lined with rigidized stainless steel up to the top of the main window line (i.e. to the top of the window post caps). Elsewhere the body interior including sides roof, and front and rear ends shall be fully lined with metal or other substantial puncture-resisting, hard surface lining covered with supported vinyl plastic, melamine

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plastic sheets or combination thereof. The material shall be washable and shall blend with the interior finish of the bus. Interior metal trim shall be anodized aluminum or stainless steel.

* 3.5.6 Windshield. A safety glass windshield affording maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body. The glass shall be tinted.

3.5.7 Windows. The main windows shall be of a size and an arrangement suitable for the intended service and shall provide maximum practicable vision for the seated passengers. The main windows shall be of the horizontal sliding type with anodized aluminum or stainless steel frame, and with finger-tip control permitting ready opening and adjustment to various desired openings. The main windows shall provide a clear opening of not less than 29 inches by 30 inches and shall be of adequate size to permit unloading litter patients through the windows in an emergency. Stops to restrict the horizontal sliding action of the windows shall be furnished and shall be set to provide an opening of approximately (but not exceeding) six inches. Windows shall be rattleproof and, when closed, rainproof, waterproof, and windproof. All windows shall be fitted with locking devices. Driver's window(s) shall permit use for signaling and for ventilation.

3.5.7.1 Windows for litter patients. The window arrangement shall also permit outside vision by litter patients in all litter tiers. Fixed windows shall be furnished above and below the main windows, where required, to provide visibility.

* 3.5.7.2 Tinted windows. Tinted, heat absorbing, safety window glass shall be furnished. Entrance door windows, driver's left window(s) and windshield shall be in accordance with the manufacturer(s) standard for tinted glass. Tinting of lamination is not acceptable except for dark 6-inch bands at the top of the driver's and entrance windows, if manufacturer's standard. Side windows in passenger area shall be tinted to double density or shall be of the graduated tinted type or shall be tinted to allow no more than 33 percent light transmission. Rear window(s) shall be tinted to double density allowing not more than 33 percent light transmission or a solid insulated nontransparent panel shall be furnished. For passenger area side and rear windows, tinted glass with tinted lamination may be used.

3.5.8 Seating. Seating shall comply with Federal Motor Carrier Safety Regulation 393.91. Deluxe intercity-type seating shall be provided to accommodate 40 passengers and a driver. All seats shall be of the forward-facing type on not less than 33.5 inch centers. Ten two-passenger, individual type, reclining-type seats shall be provided on each side with the

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front seat cushion spaced at least 12 inches from the modesty panel and 9 inches from the back of the driver's seat. All seats, except for the four attendants' seats, shall be readily removable from the bus for rapid conversion of the vehicle for carrying litter patients. All seats shall be secured to prevent shifting under all operating conditions. All mounting hardware shall be corrosion resistant and reusable and shall be quick acting but not operable by hand, to insure that they cannot be released by unauthorized persons. The seats shall be recess-mounted to the floor. Recessed track to permit varied arrangement of seats shall be furnished. Plastic fillers covered with bonded floor covering material shall be furnished to close voids in the floor surface and to provide a blended and smooth surface when seats, except attendants seats, are removed for litter service.

3.5.8.1 Passenger seats. Frames of the passenger seats shall be constructed of formed metal tubing, wall mounted with legs on the aisle side (or with legs on both sides) and shall include armrests and two-position footrests. Seats shall be manufacturer's standard 36-inch width and shall be readily removable. Seat cushions shall be not less than 17 inches in depth from a vertical plane at the front edge to a vertical plane at the most forward part of the seat back cushion. Seat backs and seat cushions shall be individually constructed with full depth foam rubber, polyester foam, or polyurethane foam, and shall include a headrest. Cushioning materials shall be suitably fire resistant. The top, the upper corners and not less than the upper 12 inches of the rear surface of each seat back shall be constructed of or covered with force distributing material. The material shall minimize force and spread the area of contact upon impact by an occupant in the next seat to the rear. The rear surface of the seat back below the force distributing material shall be covered with stainless steel. Closure type ash receptacles shall be provided for each seating space, located in the arm rests.

* 3.5.8.2 Seat covering. Seat covering material for the passenger and driver's seat shall be combination woven fabric and vinyl plastic. Seat fabric shall be not less than 42 ounce weight per linear yard. The top of the seat cushion and the front of the seat back shall be covered with woven materials of identical pattern and design, except the force distributing material shall be covered with vinyl. Side boxing of seat cushions and seat shall be covered with vinyl material of a coordinated color with the fabric material utilized. The headrest and force distributing material shall be covered with vinyl plastic. Seat and back covering shall be readily removable. Seat covers shall be readily removable without taking armrests apart and shall be able to withstand dry cleaning. Samples of proposed upholstery material combinations shall be submitted to procuring activity for customer selection and approval.

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3.5.8.3 Tour director's seats. Two tour director's seats shall be provided for attendants use to enable them to rest and observe patients during travel. The seats shall measure approximately 15 inches by 15 inches, shall be positioned to allow the attendants to view the patients and shall be hinged to raise flush against the paneling when not in use. One seat shall be hinged to the panel to the right of the driver and one seat shall be installed in the lavatory-baggage area.

* 3.5.8.4 Driver's seat. The driver's seat shall be heavy duty, bus-type, pedestal mounted and adjustable fore-and-aft and for height without the use of tools. Seat cushions, padded with foam rubber, polyester foam or polyurethane foam, shall be provided with tilting adjustments for cushion and back to provide maximum driver comfort. All these adjustments shall be capable of being made manually by the driver while he is seated and shall be safeguarded against pinching of fingers. The adjustment mechanism shall be sufficiently rugged to hold adjustment in service, shall be properly shielded from dirt, shall be located so as not to be a hazard during operations and shall preclude accidental raising or lowering. The seat back shall be covered with stainless steel or scuff and impact resistant plastic. A lap type seat belt shall be installed for the driver's seat.

3.5.9 Stanchions and grab rails. Stanchions and grab rails of stainless steel, stainless-clad steel, or plastic-coated steel tubing, at least 1-1/4 inches in diameter, shall be installed. The grab rails in the service door area shall be not less than 1 inch in diameter. A slide or roll type opaque curtain to provide protection for the driver from passenger interference shall be installed. A complete modesty panel behind the driver or a closure panel to the right of the driver shall be provided to prevent right front aisle passenger from accidentally disengaging the gear shift. A stanchion and grab rail with modesty panel shall be provided at the rear of the service door entrance and a stanchion or grab rail shall be provided at the front of the entrance, arranged to assist passengers in entering or leaving the bus. Two overhead handrails shall be provided, one on each side of the center of the aisle, for aiding the attendant in the treatment of patients. The handrails shall be at the lowest practicable height from the floor, but shall not interfere with placing a litter patient in the top tier, nor project below the minimum clear height of 79 inches in the aisle specified in 3.5.1.

3.5.10 Heating, defrosting, ventilating and cooling. Heating, defrosting. and ventilating systems shall conform to Federal Motor Carrier Safety Regulations 393.77 and 393.79.

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3.5.10.1 Heater. The main system shall provide filtered, fresh, and recirculated airflow through a common duct system. The duct system shall disperse air through louvered vents below the sash of all side windows or through fin tube ducting installed at the ceiling center or at the junction of the floor and both side walls, and extending the entire length of the bus passenger area. A dash system shall furnish heat in the area of the driver, windshield, and door. Engine-coolant hot water type heater(s) shall be provided. The capacity of the heating system shall maintain the inside bus temperature at a minimum of plus 70°F, midway between floor and ceiling, in ambient outside air of minus l0°F, with no passengers inside and air control at 100 percent recirculation. The temperature gradient between floor and ceiling shall not exceed 10°F. Blower capacities shall provide a ventilating airflow of at least 1200 cubic feet per minute (cfm). The system shall be provided with thermostatic controls. The system shall also be capable of providing 20 percent fresh air and 80 percent recirculated air or 100 percent recirculated air with controls for proportioning the amount of fresh air from 0 to 20 percent.

* 3.5.10.2 Defroster. Defroster outlets with suitable connections and independent blower(s) shall be furnished. The independent blower(s) shall have a rating of not less than 250 cfm, and shall be installed to provide a flow of heated air to the full windshield at not less than 30,000 British thermal units (Btu) per hour. An adjustable auxiliary fan shall be provided on top of the dash with a guard. The windshield defrosting system shall conform to SAE J381 and SAE J382.

3.5.10.3 Ventilator. A manual, adjustable, flow-type ventilator opening shall be provided at the front of the bus. The outside openings for heaters and ventilators shall be so located as to minimize entry of fumes, road dust, rain, or road wash and shall be covered in such a way as to prevent clogging in bad weather.

* 3.5.10.4 Air conditioning. The vehicle shall be equipped with the manufacturer's standard air conditioning system. The manufacturer shall furnish to the engineering department of the procuring activity the rated capacity at the specified conditions of the entire unit, either by air measurement or the flow meter method. The air conditioner shall have a cooling capacity of not less than 72,000 Btu per hour and shall be capable of maintaining not more than 80°F dry bulb and a 68°F wet bulb temperature and an effective temperature of 74°F inside the bus when the ambient temperature is 95°F dry bulb. The humidity shall be not less than 30 percent and not more than 60 percent. Inside temperature shall be measured midway between floor and ceiling at the front, middle, and rear seats with no passengers in bus and not more than 6°F wet bulb

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temperature gradient within the bus. The cooling performance specified herein shall be met when bus is traveling at 65 mph and when standing still with main engine at a fast idle. The air conditioning system shall be integrated with the heating system to the maximum extent possible, and shall automatically proportion air in the ratio of 20 percent fresh air, 80 percent recirculated air. All air shall pass through the filter system. Air conditioner controls shall be panel mounted within easy reach of the seated driver. The air conditioner compressor and fans shall be driven by other than a hydraulic system.

* 3.5.10.4.1 Independent system. When an independent air conditioning system is furnished, it shall be powered by a diesel engine of not less than 35 horsepower rating. Engine fuel shall be from the main engine fuel tank. All air conditioning components except the evaporator shall be located in a compartment under the floor. Air conditioning engine starter safety control switches to cut out driver's controls, compartment lamp switch and junction panel shall be installed in the auxiliary engine compartment, in addition to controls provided at the driver's seat. The air conditioner shall be designed for bus operation and shall be complete with all necessary controls for automatic operation of the unit after the power unit has been started and the compressor engaged. The air conditioning system may include a 110-volt ac alternator, driven by the auxiliary engine, to power air conditioning blowers.

3.5.11 Bus body equipment. Body equipment shall be complete with all accessories listed by the bus body manufacturer as standard equipment and, in addition, shall include items specified in 3.5.11.1 through 3.5.11.10.

* 3.5.11.1 Rearview mirrors. One nonglare type mirror having not less than 49 square inches of reflective area or a convex mirror not less than 35 square inches in area shall be mounted inside the bus. Two rearview mirrors shall be mounted outside at the front of the bus. The reflecting surface of each outside mirror shall be not less than 8 inches by 8 inches in size. Outside mirrors shall be mounted one each on the left and right side of the bus and shall have retractable arms. Mirrors shall be friction mounted to prevent damage when struck. Additional mirrors as required shall be provided for adequate driver's viewing the loading of ambulatory patients and passengers.

3.5.11.2 Sunvisor. An adjustable sunvisor shall be furnished.

* 3.5.11.3 Emergency reflective triangles. Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i) shall be furnished and properly stowed.

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* 3.5.11.4 Emergency equipment. Emergency equipment shall include first aid kit provisions, a small fire axe, and a fire extinguisher conforming to Federal Motor Carrier Safety Regulation 393.95(a). Emergency equipment shall be installed near the front of the coach convenient to the driver. The fire extinguisher shall be a minimum 2-1/2 pound, dry chemical type; shall have an Underwriter's Laboratory rating of 5B:C or more; shall include a metal safety pin and wire; and shall conform to type I or type II, class 2, size 2-1/2 of O-E-915. The fire extinguisher shall permit visual determination as to whether it is fully charged. Not less than a 9-unit type first aid kit shall be furnished or a bracket shall be provided for future installation of a 12 unit type first aid kit. The 12-unit type first aid kit size is 9-7/16 inches long, 5-7/16 inches wide, and 2-7/8 inches thick. The 12-unit type kit is not required to be furnished with the vehicle.

3.5.11.5 Stowage compartment(s). Provision shall be made for weatherproof stowage of tools and tire chains under lock and key.

3.5.11.6 Rear bumper. The rear bumper shall have anti-ride and anti-hitch features and shall permit the bus to be pushed without permanent distortion of bumper, body, or chassis parts.

3.5.11.7 Keys. Duplicate keys shall be furnished for all key-operated locks. Keys shall be identified and secured to vehicle steering wheel or otherwise stowed in an approved manner for shipment. All locks on the same vehicle shall be of the same key code.

3.5.11.8 Standee line. Areas prohibited to standees shall be marked in accordance with Federal Motor Carrier Safety Regulation 393.90.

* 3.5.11.9 Baggage compartment. When a mid-mounted engine is furnished, the baggage compartment(s) shall provide not less than 125 cubic feet. When a rear engine is furnished, the baggage compartment(s) shall provide not less than 212 cubic feet of usable space. All baggage compartments shall be adequately illuminated, shall incorporate a switch, and shall be provided with means of secure locking.

3.5.11.10 Toilet and lavatory. A standard intercity bus type lavatory and toilet facilities shall be provided at the rear of the vehicle. The facilities shall include a flush or chemical toilet, wash basin with hot and cold water, toilet tissue and paper toweling dispensers, 12-volt light fixture(s) with switch, adequate water tank with exterior water inlet, exterior sewage drain outlet, power vent and hand rails designed to aid disabled patients, and a full modesty pull curtain in lieu

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of a door. Facilities shall have capacity to support the full load of 40 passengers plus driver for not less than 4 hours continuous travel. Exterior drain and water inlet connections shall be indexed so that the drain hose cannot be connected to the water inlet, and the connections shall be clearly marked for identification.

3.5.12 Ambulance equipment. The vehicle shall be furnished with additional equipment for the care and treatment of litter and ambulatory patients as specified herein.

3.5.12.1 Litter kit. The bus shall accommodate not less than 18 litters (FSN 6530-783-7205, Litter Folding, Folding Pole, and FSN 6530-783-7905, Litter, Folding Rigid Pole) suspended in three tiers along each side of the bus and four seated attendants or shall be converted to seat not less than 36 ambulatory patients and four seated attendants. The vertical clearance between litters measured from center to center of litter poles and between the top litters and the roof shall be not less than 22 inches (26 inches desirable) to allow space to render treatment of litter patients. Nominal dimensions of litters are 90 inches long, by 22-3/4 inches wide and 6-1/2 inches high. Litter support installation shall provide not less than 6 inches from ends of litters to front attendants seat(s); not less than 14 inches from nearest rearward vertical obstruction; and shall have not less than 8 inches between handle ends of remaining litters with approximately equal spacing at each end of the central bank of litters. The installation shall secure litters effectively without discomfort to patients. Litter supports shall be furnished as specified. Web straps, chains and complicated holding straps will not be acceptable. Adjusting nuts on the litter retention devices shall be captive.

3.5.12.1.1 Additional design and construction criteria. All litter supports and attachments shall be adequate for safe support of 250-pound patients on the litters under all static and mobile operating conditions specified herein or resulting from normal operation. Litter support installation shall effectively reduce lateral or longitudinal sway of the occupied litters during normal acceleration, deceleration and turning. Use of shock absorbing support members for limiting transmission of road shock to the litter patients must be considered in the design and utilized to the maximum practicable extent.

* 3.5.12.1.2 Litter supports. Litter support brackets shall be of the cantilever type mounted on support posts so as to be readily removable. Support posts shall fold away against the side of the vehicle. They shall be of aluminum alloy construction in accordance with Douglas drawing No. 5928496-1. Each bracket shall have two built-in channels with attached

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quick-engagement-type clamps or latches to retain the litter poles; with a bracket provided at both ends of each litter. Brackets shall be designed to prevent litters from shifting forward of rearward. Sufficient space shall be provided between the bracket and the wall to provide hand clearance. Brackets and posts shall be removable. Installation shall positively secure and support 250 pounds per litter under all dynamic conditions. Litter support posts shall be constructed in accordance with Douglas stanchion number 5933765-1 with necessary matching hardware to secure post assembly to the floor and sidewall of the vehicle. Installation shall limit transmission of road shock to the litters; and shall allow ready insertion and removal of any number of loaded litters up to and including full vehicle litter capacity. Any litter support hardware which is not readily removable shall not interfere with seat placement, seat adjustment or the seated passengers.

3.5.12.2 Hanger stowage. A compartment shall be provided for stowage of litter supports which permits and maintains stowage of supports in an orderly manner. Additionally, the compartment shall be adequate to maintain orderly stowage of securing straps and belts specified below.

3.5.12.3 Securing straps. Permanent type hooks with detachable straps or belts shall be provided at at least two locations on each vehicle side wall to secure folded wheel chairs and stryker frames. The straps shall be equipped with a fast release mechanism. The hooks shall be recessed in the wall when not in use.

3.5.12.3.1 Occupied wheelchairs. Provision shall be made for securing to the seat tracks at least two occupied wheelchairs on each side of the bus. Wheelchairs shall be retained in the area normally occupied by the last bank of litters or the last three rows of seats. All devices shall be removable or retractable when not in use so as not to interfere with use of either litter or seats.

* 3.5.12.4 Container hooks. Hooks for suspending intravenous solution containers, either plastic bags or bottles, shall be attached off center of the aisle at the ceiling every 36 inches from front to rear of the ambulance. The container hooks shall be recessed into the ceiling when not in use. Hooks shall be of the removable type.

3.5.12.5 Medical cabinet. A medical cabinet, not less than 36 inches long by 18 inches high and 12 inches deep, shall be provided for storage of equipment such as resuscitator, oxygen and other medical supplies. The cabinet shall be equipped with positive latch(es) to secure door(s) during transit. The cabinet shall be completely lined with stainless steel.

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3.5.12.6 Liquid dispensers. The bus shall be equipped with dispensers for hot and cold liquids. The dispensers shall be two units of FSN 7330-781-3859, Jug, Vacuum, CRES Outer Shell, CRES Inner Vessel; with gasketed metal cover and snap locking device closure; with spigot assembly; in accordance with MIL-J-20349 Style B, 5-gallon capacity. The mounting in the bus shall be at convenient locations and heights for dispensing drinks, and shall permit ready removal of the jugs for filling, emptying, and cleaning.

3.5.12.7 Radio and intercom. An AM-FM radio shall be provided together with 8 speakers installed in the ceiling center and with a speaker in the driver's compartment. Each speaker shall incorporate an on-off switch. Additionally,. a 3-way on-off switch for the radio system shall be installed in the driver's compartment and at the rear of the coach. One wall mounted intercom station with retractable phone piece shall be mounted on the side or cabinet wall near the rear of the coach for attendants working in the lavatory-baggage area. An identical intercom station shall be installed in the driver's compartment in such a location as to provide easy access to the driver. The system shall permit transmission and reception from both stations. The radio and intercom shall operate satisfactorily with all equipment in operation.

3.5.12.8 Vacuum outlet, medical. A plug-in or screw type, self-sealing valve connection shall be installed in the patient compartment, one between each pair of vertical tiers of litters on each side of the bus (total of 4 valve connections). The suction lines shall be connected to the valve mechanisms, using the vacuum created by the vacuum pump. The units shall prevent the entry of dust without the use of a dust cap or cover. Secondary equipment attachment and release shall be capable of being performed with one hand. The outlets shall prevent twist or turn of the secondary equipment, once the equipment is connected. A mating male fitting shall be provided for each outlet. The vacuum system shall include a vacuum pump capable of pulling at least 2 standard cubic feet per minute and shall further incorporate a vacuum reservoir tank and regulator with a capacity of at least 4.27 cubic feet. The pump shall have the capability of pulling a vacuum of 29 inches of mercury.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government, contractor shall service and adjust the vehicle for operational use including at least the following: alinement of lights; adjustment of the main and auxiliary engines, electrical and brake system; filling and charging of battery; alinement of front wheels; inflation of all tires; complete

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lubrication of body, chassis, engines, and running gear, with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol type antifreeze and water in equal parts by volume; and servicing of windshield washer reservoir with water and appropriate additives.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Radio frequency suppression verification. The vehicle manufacturer shall furnish with the questionnaire (commercial vehicle engineering data) a description of all interference suppression devices used to suppress the vehicle electromagnetic radiation in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle.

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4.3.2 Vehicle weight. The vehicle, with seats installed and fully equipped and furnished, shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles will be computed using the curb weight and the payload, uniformly distributed over the vehicle load area, to provide the GVW as specified in 3.2.5. Calculated imposed loads on front and rear axles will be utilized to ascertain that suspension, axles, and tires furnished are of adequate capacity to meet contractual requirements.

4.3.3 Road test. The vehicle will be examined and road tested for a distance of not less than 500 miles with payload by the contractor to assure that the vehicle will operate in accordance with the contractual requirements.

4.3.4 Noise, heating and air conditioning tests. The vehicle shall be tested to determine conformance to all noise level, heating and air conditioning requirements specified in 3.5.2.1, 3.5.10.1 and 3.5.10.4, respectively. Tests shall be as specified in 4.3.4.1 and 4.3.4.2. Test data, including a description of all applicable and pertinent test conditions, shall be recorded and furnished to the Government representatives.

4.3.4.1 Noise levels (see 3.5.2.1). Sound measurements will be made with sound measuring equipment which meets the requirements of "American National Standard Specification for Sound Level Meters," Type 2, ANSI S1.4-71; "American National Standard Specification for Octave, Half Octave, and Third Octave Band filter sets," S1.11-66; and "USA Standard Preferred Frequencies and Band Numbers for Acoustical Measurements," USAS S1.6-67.

4.3.4.1.1 Test conditions. Sound levels shall be taken at 1 foot, 3 foot and 6 foot microphone height positions at three locations in the aisle, (a) next to the front passenger seat, (b) next to the center passenger seat, and (c) next to the rear passenger seat with vehicle traveling at 30 and 55 mph in direct drive on a dry smooth level paved road with all powered accessories operating, all windows closed, and no more than four people within the vehicle. The test shall be conducted under conditions of minimal ambient noise, the level of which shall be recorded.

4.3.4.1.2 Meter calibration. The sound level meter shall have been calibrated either by the manufacturer, in the manufacturer's local office or by the contractor using the manufacturer's approved calibration equipment and techniques within a period of one day of its use in these measurements.

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4.3.4.1.3 Deviations. Any deviation from the above in noise level instrumentation, technique, test conditions, or results shall not be acceptable without prior complete information to and acceptance from the contracting officer.

4.3.4.2 Heating and air conditioning. To verify conformance to the heating and air conditioning performance requirements specified in 3.5.10.1 and 3.5.10.4, the contractor shall test the vehicle under actual or simulated conditions, at least as stringent as specified therein. Simulated conditions shall be acceptable to the Government.

4.3.4.3 Comfort evaluation. As part of the road test, the bus shall be assessed for comfort, both for litter patients and seated ambulatory patients, on straight and winding primary and secondary roads, and during acceleration, deceleration, medium speed (30 mph) and maximum authorized speed runs.

4.3.4.4 Exterior sound level. The vehicle shall be tested in accordance with SAE J366 to determine conformance to the exterior sound level requirements specified in 3.4.23.

4.3.5 Ambulance equipment test. The vehicle shall be tested to determine conformance to lift or ramp loading, litter loading, and all ambulance features and equipment requirements specified in section 3 of this specification. Tests shall include, but not be limited to, the following items.

4.3.5.1 Lift or ramp test. Test of lift loading shall include loading with ambulatory (including on crutches), wheel chair, and litter patients (simulated). Each individual shall weigh 250 pounds (including the litter bearers). The lift shall be tested for adequacy of its size, capacity, guard rails, operating smoothness, safety to attendants and patients, and performance loaded and unloaded.

4.3.5.2 Litter support test. Test of the litter support installation shall demonstrate ease of putting loaded litters in place, and minimal longitudinal or lateral movement. Strength shall be demonstrated by running the bus/ambulance over secondary roads at practical speeds with at least one vertical tier of three litters in place, each litter bearing a 250-pound simulated patient. Any breakage or distortion of the litter support installation will constitute failure.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of the vehicle until corrective action has been taken.

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4.3.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production samplet and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4 Inspection of production vehicles. Each production vehicle (less payload) shall be examined and driven on approximately level road surfaces by the contractor to assure that the vehicle will operate in accordance with the contractual requirements. The contractor may authorize his sub-contractor who will perform the final assembly work on the vehicle to make such examination and tests, but this shall not relieve the contractor of his obligation to deliver an acceptable vehicle or constitute a waiver of any warranty provision. Each vehicle shall be tested to determine that heating, air conditioning, lift platform, vacuum outlets, mechanical and electrical systems are functional. Seat removal and litter kit installation shall be checked for functional compatibility and completeness on at least one out of every five production units. Seats shall be installed in normal position for shipment.

4.4.1 Rejection. Deficiencies of workmanship and nonconformance to any requirement of the contract shall be cause for rejection until corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

* 6.1 Intended use. The vehicle covered by this specification is intended for nontactical use by the Government in the long distance transportation of ambulatory patients and conversion to transportation of litter patients. The vehicle is to be used principally in transporting patients between air terminals and hospitals and between hospitals and other points.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Identification of appropriate service for painting and marking (see 3.1.1.1).

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(c) Concealed marking, if required (see 3.1.1.1).
(d) Manual transmission, if required (see 3.4.6.2).
(e) Oil seals, if required (see 3.4.10).
(f) Traction control, if required (see 3.4.10.1).
(g) Disc type wheels, if required (se 3.4.11).
(h) If a spare wheel assembly is not required (see 3.4.11.3).
(i) If a spare tire is not required (see 3.4.11.4).
(j) If tools are not required (see 3.4.18).

* 6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction Procedure form, and computation for low speed requirement for the vehicle furnished under contract, should be submitted as specified in the contract. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and in still, dry air. The factors to be used in predicting vehicle ability (see 3.3) are established as follows for the corresponding SAE J688 Truck Ability Prediction Procedure Tables:

Table 1	- Tire factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude factor. 1.00
Table 3	- Rolling factor. 2.080
Table 4	- Area factor. 0.197
Table 5	- Velocity factor. 432.0
Table 6	- Altitude factor. 1.00
Table 7	- Chassis friction horsepower. Use applicable power unit GVW (to nearest, higher, 1000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 60 mph geared speed. For GVW and engine speed beyond the range of this table, the following formula shall be used: FHP = 1 + (0.0000002 rpm) (GVW).
Table 8	- Grade factor. 0.62
Table 8A	- Correction factor. Not required.
Table 9	- Road factor. 0.0

6.4 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (addition, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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